COURSE TITLE	**:**	**High Performance Computer Architecture**
COURSE CODE	**:**	**6151**
COURSE CATEGORY	**:**	**ELECTIVE**
PERIODS/WEEK	**:**	**4**
PERIODS/SEMESTER	**:**	**60**
CREDITS	**:**	**4**

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	**Parallel Computer Models**	15
2	**Advanced Processor and Memory Technology**	15
3	**Pipelining and Superscalar Techniques**	15
4	**Parallel And Scalable Architectures**	15

Course General Outcomes:

Module	G.O	On completion of this course the student will be able :
1	1	To Understand the concept of Parallel Computer Models
	2	To Understand the concept of Hardware and Software parallelism
2	1	To Understand the concept of Advanced Processor and Memory technology
	2	To Understand the concept of Cache and Shared Memory
3	1	To Understand the concept of Pipelining Techniques

	2	To Understand the concept of Superscalar Techniques
4	1	To Understand the concept of Multiprocessor and Multicomputer Systems
	2	To Understand the concept of Multivector and SIMD Organisation
	3	To understand the concept of Multithreaded architectures

Specific Outcomes:

MODULE I. Parallel Computer Models

1.1 To Understand Parallel Computer Models

1.1.1 To explain Flynn's Classification with diagram
1.1.2 To discuss Parallel/Vector computers
1.1.3 To describe shared memory multiprocessors
1.1.4 To explain UMA model with block diagram
1.1.5 To describe distributed memory multicomputers
1.1.6 To discuss Vector supercomputer
1.1.7 To explain architecture of vector supercomputer
1.1.8 To discuss SIMD supercomputer
1.1.9 To explain operational model of SIMD supercomputer
1.1.10 To discuss PRAM and VLSI models
1.1.11 To explain PRAM model with diagram

1.2 To Understand Hardware and Software parallelism

1.2.1 To describe conditions of parallelism
1.2.2 To discuss data dependence
1.2.3 To discuss control dependence
1.2.4 To discuss resource dependence
1.2.5 To describe hardware and software parallelism
1.2.6 To discuss hardware parallelism
1.2.7 To discuss software parallelism
1.2.8 To explain the mismatch between hardware and software parallelism

MODULE II.Advanced Processors and Memory Technology

2.1 To Understand Advanced Processors and Memory Technology

2.1.1 To differentiate CISC and RISC Architectures
2.1.2 To compare the architectural characteristics of CISC and RISC
2.1.3 To discuss superscalar architecture
2.1.4 To explain architecture of IBM RISC System/6000 superscalar processor
2.1.5 To explain VLIW architecture
2.1.6 To explain pipelining in VLIW processors
2.1.7 To discuss vector and symbolic Processors
2.1.8 To explain Vector pipelines
2.1.9 To explain symbolic processors
2.1.10 To explain architecture of the Symbolic 6000 Lisp Processor
2.1.11 To discuss four level Memory Hierarchy
2.1.12 To discuss virtual memory models
2.1.13 To explain private virtual memory
2.1.14 To explain shared virtual memory

2.2 To Understand Cache and Shared Memory

2.2.1 To discuss cache addressing models
2.2.2 To explain physical address caches
2.2.3 To explain virtual address caches
2.2.4 To discuss multi-level cache memories
2.2.5 To discuss Interleaved memory organisation
2.2.6 To explain memory interleaving
2.2.7 To explain pipelined memory access

MODULE III. Pipelining and Superscalar Techniques

3.1 To Understand different pipeline techniques

3.1.1 To discuss linear pipeline processors
3.1.2 To explain asynchronous model
3.1.3 To explain synchronous model
3.1.4 To discuss nonlinear pipeline processors
3.1.5 To explain Reservation tables and latency analysis

3.1.6 To discuss instruction pipeline design
3.1.7 To explain instruction pipeline phases
3.1.8 To explain the mechanisms for instruction pipelining
3.1.9 To discuss arithmetic pipeline design
3.1.10 To explain static arithmetic pipelines
3.1.11 To explain multifunctional arithmetic pipelines

3.2 To Understand Superscalar Pipeline Design

3.2.1 To describe superscalar pipeline design
3.2.2 To explain superscalar pipeline structure
3.2.3 To explain superscalar pipeline scheduling
3.2.4 To explain the architecture of DEC Alpha 21064 processor

MODULE IV. Parallel And Scalable Architectures

4.1 To Understand Multiprocessor and Multi computers

4.1.1 To describe hierarchical bus systems
4.1.2 To explain Ultrmax multiprocessor architecture
4.1.3 To discuss multi computer architectures
4.1.4 To explain node architecture of multicomputer
4.1.5 To explain router architecture of multicomputer

4.2 To Understand Multivector and SIMD Computers

4.2.1 To discuss vector processing principles
4.2.2 To explain NEC SX-X 44 vector supercomputer architecture
4.2.3 To explain distributed memory SIMD system model
4.2.4 To explain shared memory SIMD system model

4.3 Multithreaded architecture

4.3.1 To explain multithreading issues and solutions
4.3.2 To explain architecture of multiple context processor I
4.3.3 To discuss multidimensional architecture

<u>CONTENT DETAILS</u>

MODULE I - **Parallel Computer Models**

Parallel Computer Models :: Flynn's Classification-Parallel/Vector Computers-Multiprocessors and multicomputers-Shared memory multiprocessors-UMA Model-Distributed memory multicomputers-Vector supercomputer- architecture of vector supercomputer-SIMD supercomputer-Operational model-PRAM and VLSI models.

Hardware and Software Parallelism :: Conditions of parallelism-data dependency and resource dependency-hardware and software parallelism-mismatch between Hardware and software parallelism.

MODULE II – **Advanced Processor and Memory Technology**

Advanced processor technology::CISC and RISC scalar processors—Superscalar architecture -VLIW architecture-pipelining in VLIW processors-Vector and symbolic processors-vector pipelines-Memory hierarchy technology-Virtual memory models-private and shared virtual memory.

Cache and Shared Memory :: -Cache memory organization-Cache addressing models-Physical address caches-Virtual address caches-multi-level cache memories. Shared memory organization-interleaved memory organization-memory interleaving -pipelined memory access.

MODULE III – **Pipelining and Superscalar Techniques**

Pipelining Techniques :: Linear pipeline processors- Asynchronous & Synchronous models-Nonlinear pipeline processors-reservation tables and latency analysis -Instruction pipelining design-Instruction execution phases-Mechanism for instruction pipeline-Arithmetic pipelining -Static arithmetic pipelines -Multifunctional arithmetic pipelines.

Superscalar Techniques:: Superscalar pipeline design-Super scalar structure-superscalar pipeline scheduling-superscalar architecture.

MODULE IV – **Parallel And Scalable Architectures**

Multiprocessors And Multicomputers:: Hierarchical bus systems-Multiprocessor architecture-multicomputer architecture –node and router architecture.

Multivector and SIMD Computers:: -Vector supercomputer architecture. SIMD Computer organizations-Distributed memory model- Shared memory model.

Multithreading processors:: Multithreaded architecture-multithreading issues and solutions-multiple context processor-multidimensional architecture.

Text Book(s):

1 . Advanced Computer Architecture – KAI HWANG & NARESH JOTWANI – Tata McGraw Hill Edn. Second Edition

REFERENCES:

1. John L. Hennessey and David A. Patterson, " Computer Architecture – A quantitative approach", Morgan Kaufmann / Elsevier, 4th. edition, 2007.
2. David E. Culler, Jaswinder Pal Singh, "Parallel Computing Architecture : A hardware/ software approach" , Morgan Kaufmann / Elsevier, 1997